Jae Ho Chang, Ph.D.

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Professional Experience

Postdoctoral Researcher 2026–
Yale University Connecticut, USA

† Faculty mentor: Dr. Yize Zhao

† Fields of study: Biostatistics, mental health

Education

Ph.D. in Statistics 2021–2026
The Ohio State University Ohio, USA

† Dissertation: Methodological Advances in Network Inference and Information Transfer

† Advisor: Dr. Subhadeep Paul, Dr. Arnab Auddy (co-advisor)

B.A. & M.A. in Applied Statistics 2012–2020
Konkuk University Seoul, ROK

Research Papers

In Progress

‡ **Chang, J.**, Auddy, A., and Paul, S. (2025–). Transfer learning for graphical models and differential graph estimation with FDR control.

‡ Residual Learning Enhanced Koopman Model Predictive Control for Unmanned Aerial Vehicles.

Under Review

‡ Vadlamani, A., **Chang, J.**, and Parthasarathy, S. (2026). Transfer Calibrated Prediction Powered Inference. *arXiv preprint*.

‡ **Chang, J.**, Auddy, A., and Paul, S. (2026). Personalized Federated Learning via Variance-Aware Nonparametric Empirical Bayes. *arXiv preprint*.

‡ Nath, S., Hong, J., **Chang, J.**, Warren, K. and Paul, S. (2025). Recidivism and Peer Influence with LLM Text Embeddings in Low Security Correctional Facilities. *Journal of Econometrics*. Revise and resubmit.

‡ **Chang, J.**, Russo, M. and Paul, S. (2024). Heterogeneous Transfer Learning for High-Dimensional Regression with Feature Mismatch. *arXiv preprint*.

Published

‡ **Chang, J.** and Paul, S. (2026). Embedding Network Autoregression for Time Series Analysis and Causal Peer Effect Inference. *Journal of Machine Learning Research*, 27(111):1–72, 2026.

‡ **Chang, J.**, Lee, S., and Kim, H. (2023). An Asymptotically Efficient Closed-Form Estimator for the Dirichlet Distribution. *Stat*, 12(1).

‡ Lee, S., **Chang, J.**, and Kim, H. (2021). Further Sharpening of Jensen's Inequality. *Statistics*, 55(5).

‡ **Chang, J.**, Moon, H., and Kwon, S. (2021). High-Dimensional Linear Discriminant Analysis with Moderately Clipped LASSO. *Communications for Statistical Applications and Methods*, 28(1).

Awards and Honors

Best Student Poster Awards May 2026
Among top 5 out of 20, The 11th Workshop on Biostatistics and Bioinformatics Georgia, USA
American Statistical Association (ASA) Georgia Chapter

D. Ransom & Marian Whitney Award for Outstanding Research Associate May 2026
Department of Statistics, OSU Ohio, USA

Travel Awards 2025–2026
Conferences and Workshops USA

† Workshop on Collaborative Research at the Intersection of Statistics and Engineering (CRSE), Wharton School, University of Pennsylvania

† Workshop on New Horizons on Model Transportability and Data Integration, Institute for Mathematical and Statistical Innovation (IMSI)

† The 11th Workshop on Biostatistics and Bioinformatics, Georgia State University

† Annual Winter Workshop on Frontiers in Learning Under Data Heterogeneity, University of Florida

† The Inaugural Workshop on Frontiers in Statistical Machine Learning (FSML), Institute of Mathematical Statistics (IMS)

Selected Presentations

Residual Learning Enhanced Koopman Model Predictive Control for UAVs

† Talk, CRSE Workshop, Wharton School at University of Pennsylvania — Aug 2026

Personalized Federated Learning via Variance-Aware Nonparametric Empirical Bayes

† Poster, CRSE Workshop, Wharton School at University of Pennsylvania — Aug 2026

† Talk, Transfer Learning through Empirical Risk Minimization, JSM — Aug 2026

† Poster, New Horizons on Model Transportability and Data Integration, IMSI — Jun 2026

† Poster, The 11th Workshop on Biostatistics and Bioinformatics, Georgia State University — May 2026

† Poster, 2026 Annual Joint Biostatistics Symposium, OSU — Jan 2026

Heterogeneous Transfer Learning for High-Dimensional Regression with Feature Mismatch

† Poster, Statistics Annual Winter Workshop, University of Florida — Jan 2026

† Talk, Advances in Transfer Learning, JSM — Aug 2025

† Poster, IMS Inaugural FSML Workshop, Vanderbilt University — Aug 2025

Embedding Network Autoregression for Time Series Analysis and Causal Peer Effect Inference

† Talk, Virtual Workshop for Junior Researchers in Time Series — Mar 2026

Selected Projects

Heterogeneous Transfer and Personalized Federated Learning Methods for Digital Twin in Unmanned Aerial Vehicles

Sep 2025–Aug 2026, USA

NSF Award 2529302 – $570K (PI: Subhadeep Paul)

† Worked as a graduate researcher developing statistical and algorithmic foundations for heterogeneous transfer learning and personalized federated learning with aerodynamic digital twin applications.

Transfer, Federated, and Private Statistical Learning for Trustworthy AI

2024–2025, USA

OSU College of Arts and Sciences (NMS) Exploration Grants Program

† Worked as a graduate researcher developing heterogeneous transfer learning for high-dimensional regression with feature mismatch.

Predictions of Allogrooming in the Vampire Bat Social Network

May–Aug 2024, USA

Collaboration with Department of EEOB, OSU

† Investigated social bonding and cooperative behavior in vampire bats using network analysis to predict allogrooming patterns.

Antibiotic Stewardship System Development

Jun–Sep 2018, ROK

SK C&C, Korea University Medical Center

† Built classification models for predicting methicillin-resistant Staphylococcus aureus (MRSA) and vancomycin-resistant enterococci (VRE).

† Developed APIs for clinical deployment of predictive models.

Professional Affiliations and Service

Professional Affiliations

† Member of ASA and IMS

Peer Review

- † 2026: Journal of the Royal Statistical Society: Series C, Statistics & Probability Letters
- † 2025: Computational Statistics & Data Analysis
- † 2021: Communications for Statistical Applications and Methods

Teaching Experience

Graduate Teaching Associate 2021–2026
Department of Statistics, OSU Ohio, USA

- † STAT 4201, STAT 4202: Introduction to Mathematical Statistics I & II (AU 22, SP 23)
- † STAT 3202: Introduction to Statistical Inference (SU 23, AU 23, SP 24, SP 25)
- † STAT 1350: Elementary Statistics (SP 22)

Workshop Lecturer Jul–Oct 2020
Department of Human Resources, GS E&C Seoul, ROK

- † Delivered tutorials on SPSS and introductory statistics, including probability and linear regression.

Graduate Teaching Assistant 2018–2019
Department of Applied Statistics, Konkuk University Seoul, ROK

- † Supported courses in Regression Analysis, Likelihood Theory, Distribution Theory, Nonparametric Estimation, and Mathematical Statistics.

Technical Skills

- † R and Python (advanced), C++ (intermediate)
- † Git, Unix, SLURM-based HPC workflows
- † LaTeX, SQL